

From Manual Real Estate Operations to Automated Lead Management

We transformed a fragmented real estate sales process into a connected workflow that captures leads, routes enquiries, automates communication and simplifies follow-ups.

Client: Trade ghar

Industry: Real Estate

Duration: 8 Weeks

Services: Workflow Automation, AI/LLM Implementation, Web Development, API Integrations

The Challenge

The business was losing time to manual operations.

The client managed property enquiries across WhatsApp groups, property portals (99acres, MagicBricks), Meta/Instagram ads, phone calls and referrals. Every lead required manual data entry, manual assignment and manual follow-up. Nothing was connected.

The old workflow:

1. Lead arrives via WhatsApp group, portal or call
2. Salesperson reads and interprets the message
3. Copies details into spreadsheet or CRM
4. Manually assigns to a team member
5. Contacts the customer
6. Remembers to follow up
7. Updates status manually

Problems:

- **Manual data entry** — Every enquiry required someone to read, interpret and copy information into a system by hand.
- **Duplicate records** — The same customer entered through multiple WhatsApp groups and channels, creating conflicting records.

- **Scattered information** — Customer data lived in WhatsApp chats, spreadsheets and personal notebooks simultaneously.
 - **No sales visibility** — Management had no real-time view of the pipeline, team activity or lead status.
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Before: A Fragmented Sales Process

Multiple lead sources fed into manual processing, producing inconsistent data spread across disconnected systems.

Lead Sources:

- WhatsApp broker groups
- Website contact forms
- Property portals (99acres, MagicBricks)
- Meta / Instagram ads
- Phone calls
- Referrals

Where Data Ended Up:

- Personal WhatsApp chats
 - Google Sheets
 - Partially-used CRM
 - Written notes
 - Group messages
 - Memory
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The Solution

We connected the entire workflow.

We designed an automated system that ingests WhatsApp broker messages via webhook, classifies them using AI, extracts structured property data, deduplicates records, creates listings, notifies relevant parties and tracks everything through a centralized dashboard.

The new workflow:

1. Lead source (WhatsApp group, portal, web form)
2. Message received via Green API webhook

3. Deduplicated against 7-day Redis cache
 4. AI pipeline classifies message into one of 7 categories
 5. Property data extracted (location, price, BHK, furnishing, owner type)
 6. Resolved against master data (locations, societies, property types)
 7. Customer record created or matched by phone number
 8. Listing persisted to database with auto-generated property ID
 9. Low-confidence results routed to admin review inbox
 10. Lead tracked in centralized admin dashboard
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What We Automated

01 — WhatsApp Message Ingestion

A standalone webhook service receives incoming WhatsApp messages via Green API, filtering to group chats. Handles 7 webhook types including text, media, location and contact messages. Two-layer deduplication: SHA-256 checksum in Redis with MongoDB fallback.

02 — AI-Powered Message Classification

Each message is classified by a Gemini LLM pipeline into one of 7 categories: Rental Owner, Rental Tenant, Seller, Buyer, Bank Auction, New Project or CoWorking. The AI uses a domain-specific persona prompt trained on Gurugram real estate terminology. Automatic failover across 4 Gemini models on quota exhaustion.

03 — Intelligent Data Extraction

A 9-phase parser pipeline combines deterministic regex with LLM-powered extraction. Location, property type, budget, BHK configuration, furnishing status, owner profile and amenities are extracted from unstructured Hinglish text. A price sanity guard catches per-sqft rates leaking into total price fields.

04 — Fuzzy Location Resolution

City-aware location resolution handles 8 satellite towns around Gurugram. Levenshtein-based fuzzy matching tolerates typos while preserving digit identity (Sector 57 never matches Sector 75). Unknown societies are auto-enriched via Google Search grounding for sector, coordinates, builder and amenities.

05 — AI-Powered Property Search

A multi-turn conversational chat interface lets customers search inventory using natural language. The system tracks conversation state across messages, expands corridor searches

(Golf Course Road expands to Sectors 42, 43, 53, 54), detects info questions about metro distances and schools, and provides zero-results fallback with 4km geo-radius expansion.

06 — Centralized Admin Dashboard

A full admin dashboard with 7 property management modules, parser inbox for reviewing AI-classified messages, customer-to-property matching engine, lead generation tracking, requirement form management, and 17 master data tables. Role-based permissions control access across the team.

The Parser Pipeline — 9 Phases

This is the core automation engine. Each WhatsApp message passes through 9 sequential processing phases:

Phase 1 — Polling A background job polls every 10 seconds, claims pending messages atomically, and releases stale claims after 2 minutes.

Phase 2 — Preprocessing Cleans WhatsApp markdown formatting. Extracts metadata: sender phone, sender name, chat name, chat ID.

Phase 3 — Classification Calls Gemini LLM with a domain-specific system prompt and classification keyword hints. Validates output against 7 property categories. Price sanity guard catches misclassified per-sqft rates.

Phase 4 — Mapping Maps LLM output to model-specific documents. Handles date coercion, money normalization (lakh/crore/k parsing), boolean flag resolution.

Phase 5 — Resolution Resolves text values to master database ObjectIds for Location, Society, PropertyType, Furnishing, OwnerProfile. Auto-creates missing locations and societies. Upserts Customer record from WhatsApp metadata.

Phase 6 — Enrichment Google Search grounding via Gemini enriches unknown societies with sector, coordinates, builder and amenities. Location centroids resolved from a human-verified coordinate reference table with Gemini fallback.

Phase 7 — Persistence Deduplication check (same sender, same group, same text within 7 days). Writes to the appropriate collection. Assigns sequential property ID via Counter pre-save hook. Syncs amenities to society documents.

Phase 8 — Image Linking Links follow-up media-only messages to listings created by the preceding text message. Handles both media-after-text (10-min forward window) and media-before-text (3-min backward window) patterns.

Phase 9 — Review Gate Routes low-confidence, unknown-category, truncated or missing-propertyType listings to the admin ParserInbox for manual review and correction.

Automation Logic

The system follows predefined business rules. Every automated decision is based on configurable rules that reflect how the business actually operates.

IF message contains property listing → Classify as supply (RentalOwner or Seller) → Extract location, price, BHK, furnishing, owner type → Create structured listing with auto-generated property ID

IF message contains requirement → Classify as demand (Buyer or RentalTenant) → Extract budget, preferred location, BHK preference → Match against live inventory

IF location is outside Gurugram region → Flag for manual review → Route to admin parser inbox

IF duplicate message detected → Check SHA-256 checksum against 7-day Redis cache → Fallback to MongoDB dedup if Redis is unavailable → Bump existing record instead of creating duplicate

IF location name has typo → Fuzzy match against known locations (Levenshtein distance) → Preserve digit identity (Sector 57 ≠ Sector 75) → Budget: 1 edit for short names, 2 for longer names → Resolve to canonical master entry

IF same city name appears in different towns → City-aware resolution (Sector 5 Sohna ≠ Sector 5 Gurugram) → Non-Gurugram towns create distinct location entries

IF parsing confidence is low → Route to admin review inbox → Allow manual correction before publishing

IF society is unknown to the system → Trigger Google Search grounding via Gemini → Auto-create society with enriched sector, coordinates, builder, amenities

IF media message follows a text listing → Link images to the preceding listing (10-min forward window) → Also check for media sent before text (3-min backward window)

AI Chat Search — How It Works

The conversational property search is a stateful, multi-turn system with several layers:

Slot Extraction: Deterministic regex extracts intent (buy/sell/rent), BHK, price ranges (handles cr/lakh/k formats), furnishing, location, society and deal flags (desperate, mandate, commission-shared, pet-friendly, brand-new). Falls back to Gemini LLM when regex confidence is low (4-second timeout).

Conversation State: Search context persists across messages. A customer can say "3BHK in Sector 57" then follow up with "under 50 lakh" — the system merges slots across turns using Gemini-powered state management.

City-Aware Search: Recognizes 6 NCR cities with typo tolerance (gurgoan, guragram, etc.). Location resolution is scoped to the detected city.

Corridor Expansion: Named roads expand to constituent sectors. "Golf Course Road" becomes Sectors 42, 43, 53, 54. "Dwarka Expressway" and "SPR" similarly expand.

Zero-Results Fallback: When no listings match, the system tries: 4km geo-radius expansion using Haversine distance, nearby-sector fallback ($N \pm 2$), budget relaxation and furnishing relaxation. Reports the cheapest available price in the area so the customer can adjust expectations.

Info Question Detection: Factual queries about metro distances, schools, hospitals, RERA and connectivity are detected and answered via Gemini with Google Search grounding — without resetting the property search context.

Slot Relaxation: Phrases like "any BHK", "anywhere" or "no budget" clear specific search slots rather than being treated as new search terms.

Automated WhatsApp Workflow

New message received via Green API webhook

Incoming (broker group): "3BHK available in Sector 57, semi furnished, 45k rent. Owner free. Contact for details."

System processes: → Dedup check: SHA-256 checksum against Redis (7-day TTL) → Parser pipeline classifies → Rental Owner Listing → Extracts: 3BHK · Sector 57 · Semi Furnished · ₹45,000/month · Owner → Resolves: Location ObjectId · PropertyType ObjectId · Furnishing ObjectId → Customer record upserted from sender phone number → Listing created with auto-generated property ID

Follow-up (2 minutes later, same sender sends photos): → Image Linker detects media within 10-min window → Photos linked to the Sector 57 listing automatically

Matching engine: → Active tenant demands for 3BHK in Sector 57 under ₹50k identified → Magic link generated with matched properties → WhatsApp notification sent to matching tenants

Customer-to-Property Matching

The matching engine connects supply and demand automatically:

- **RentalTenant** ↔ **RentalOwner** matching by location, property type, BHK, price overlap
 - **Buyer** ↔ **Seller** matching with 10% budget tolerance
 - Scoring algorithm weights: location match, property type match, bedroom match, price overlap
 - Generates magic links for customers to view matched properties
 - Magic links sent via WhatsApp template messages with automatic rotation
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Lead Generation & Contact Reveal

A gated contact reveal system manages how customers access agent information:

- **Guest visitors** must verify their phone via OTP before seeing contact details
 - **OTP system:** 6-digit codes sent via WhatsApp, bcrypt-hashed storage, 5-minute TTL, 60-second cooldown, max 5 verification attempts per 10-minute window
 - **Commission-shared properties** require additional customer verification
 - Every reveal attempt is tracked: viewer type (guest/customer), channel (view/call/WhatsApp), timestamp
 - Admin can manage lead status (new → contacted → closed), grant/revoke access
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Technical Architecture

Frontend: React, Vite, Tailwind CSS

Backend: Node.js, Express (main API)

Webhook Service: Node.js, Express (port 6060, standalone)

Database: MongoDB (30+ collections), Redis (dedup cache)

AI/LLM: Google Gemini (4-model failover chain: gemini-2.5-flash-lite → gemini-2.0-flash → gemini-2.5-flash → gemini-2.0-flash-lite)

Integrations: Green API (WhatsApp), Google Search Grounding

Infrastructure: AWS EC2, Nginx (4 subdomains with SSL), PM2 (process manager), Let's Encrypt

Authentication: OTP via WhatsApp, bcrypt-hashed, role-based permissions

Key Engineering Challenges

Unstructured Hinglish Messages

Challenge: WhatsApp broker messages arrived in free-form text mixing Hindi, English and abbreviations. No consistent format. Missing fields were common. Per-sqft rates appeared where total prices should be.

Approach: A 9-phase parser pipeline combining deterministic regex with Gemini LLM classification. A domain-specific "15-year Gurgaon broker" persona prompt improved extraction accuracy. PriceSanity guard catches per-sqft rates leaking into total price fields through statistical validation.

Outcome: Messages are parsed into structured listings with 7 distinct property categories, 30+ extracted fields including location, price, BHK, furnishing and owner profile.

Two-Layer Duplicate Detection

Challenge: The same property listing arrived from multiple WhatsApp groups simultaneously. The same broker forwarded identical messages across 10+ groups. The same customer entered through multiple channels.

Approach: Layer 1 (ingestion): SHA-256 checksum of normalized message content checked against Redis with 7-day TTL, MongoDB fallback on Redis failure. Layer 2 (parser): same sender + same group + same text window check before persistence. Customer records matched by phone number.

Outcome: Duplicate listings caught at two stages before entering the system. Duplicate counter tracks frequency (capped at 50) for analytics. Clean inventory without manual oversight.

City-Aware Location Ambiguity

Challenge: "Sector 5" in Gurugram and "Sector 5" in Sohna are different locations 40km apart. The service area covers 8 satellite towns. Users frequently misspelled society names in Hinglish.

Approach: City-scoped location resolution with Levenshtein fuzzy matching. Digit sequences preserved exactly (Sector 57 never matches Sector 75). Edit distance budget: 1 for short names, 2 for longer names. Unknown societies auto-enriched via Google Search grounding for sector, coordinates, builder and amenities. Human-verified coordinate reference table with bounding-box validation.

Outcome: Locations resolve correctly across 8 towns. Fuzzy matching handles common misspellings. Society enrichment runs automatically with Google Search grounding.

LLM Reliability at Scale

Challenge: AI classification needed to handle Gemini quota limits, model failures, inconsistent outputs and varying response latency without blocking the 10-second polling loop.

Approach: Automatic failover chain across 4 Gemini models. Temperature tuned per use case: 0.1 for parsing (consistency), 0.4 for chat (naturalness), 0 for search slot extraction (precision). Configurable retry limits per message. 2-minute stale claim release. Admin review inbox for unclassifiable messages. 4-second timeout on search LLM fallback.

Outcome: The parser runs continuously. Model quota exhaustion triggers automatic failover. Low-confidence results surfaced for human review rather than silently dropped.

Media-Text Message Ordering

Challenge: WhatsApp brokers send property details as text followed by photos, but sometimes send photos first and text second. Media and text arrive as separate messages with no explicit link.

Approach: ImageLinker phase with configurable time windows. Forward linking: photos arriving within 10 minutes after a text listing from the same sender are linked. Backward linking: media sent up to 3 minutes before the text is also captured.

Outcome: Property images are correctly associated with their listings regardless of send order, without requiring any user action.

Before and After Automation

Process	Before	After
Lead capture	Manual copy from WhatsApp	Automated via Green API webhook
Data extraction	Read and type into spreadsheet	9-phase AI parser pipeline

Lead classification	Human judgment per message	Gemini LLM with 7 categories
Duplicate handling	Manual check, often missed	Two-layer dedup (Redis + MongoDB)
Location resolution	Salesperson knowledge	Fuzzy matching across 8 towns
Customer notification	Manual WhatsApp reply	Automated via Green API
Property search	Salesperson recalls from memory	Multi-turn AI chat with natural language
Image association	Manual file management	Automatic linking within time windows
Inventory management	Multiple spreadsheets	MongoDB with 30+ collections
Matching	Manual memory-based	Algorithmic scoring with magic links
Contact gating	Open access	OTP-verified lead generation
Reporting	Manual count and summary	Real-time analytics and insights dashboard

Results

Qualitative improvements observed after deployment. Exact metrics will be measured over the first quarter of operation.

Automated — Lead capture and data entry eliminated as manual tasks across 7 property categories

Seconds — Message processing time from ingestion to structured listing, down from hours of manual work

Centralized — 30+ database collections replace spreadsheets, WhatsApp chats and notebooks

7 Categories — AI automatically classifies every message: Rental Owner, Rental Tenant, Seller, Buyer, Bank Auction, New Project, CoWorking

8 Towns — City-aware location resolution covers Gurugram and satellite towns with fuzzy matching

4-Model Failover — AI classification runs continuously with automatic model switching on quota exhaustion

Two-Layer Dedup — Duplicate listings caught at ingestion (Redis) and persistence (MongoDB) before entering inventory

Business Impact

The team stopped managing the process and started focusing on sales.

The operational burden shifted from people to software. Repetitive administrative work — reading WhatsApp messages, copying data, resolving locations, checking for duplicates, assigning leads — was replaced by a 9-phase automated pipeline.

Before: Read → Copy → Type → Assign → Remember → Follow up → Update

After: Ingest → Parse → Classify → Resolve → Dedupe → Persist → Match → Notify

The sales team now works from a centralized dashboard with clean, structured data. Customers search inventory through an AI-powered chat. Matching happens algorithmically. Contact access is gated through OTP verification. The system runs 24/7 without human intervention.

Technology and Services

Technology Stack: React · Vite · Tailwind CSS · Node.js · Express · MongoDB · Redis · Google Gemini (4-model chain) · Green API (WhatsApp) · Google Search Grounding · AWS EC2 · Nginx · PM2 · Let's Encrypt · bcrypt · OTP

Services Delivered:

- Workflow Automation (9-phase parser pipeline)
- AI / LLM Implementation (5 distinct Gemini integrations)
- WhatsApp API Integration (Green API webhooks)
- Web Application Development (customer frontend + admin dashboard)
- Conversational AI Search (multi-turn natural language)
- Customer-Property Matching Engine
- Lead Generation & Contact Gating
- Infrastructure & Deployment (AWS, Nginx, SSL, PM2)
- Process Optimization

Database Architecture

30+ MongoDB Collections:

Property Listings (7): Seller, Buyer, RentalOwner, RentalTenant, BankAuction, NewProject, CoWorking

Users & Customers: User (admin), Customer (end-user), OTP

Pipeline: WhatsappMessage, ParserInbox, ParserLog

Engagement: LeadGeneration, MatchesNotification, MagicLink, RequirementForm, ActivityLog, Testimonial

Master Data (17): State, City, Location, Society, LookingFor, PropertyType, PropertySubType, PropertyStatus, ListingStatus, ListingExpiry, Furnishing, Amenity, OwnerProfile, TenantProfile, PossessionStatus, Bank, Builder, Company, UDC, ClassificationKeyword, Counter